

Choosing the Right ML Model

Model Selection for Machine Learning



Choosing the Right ML Model

- Selecting models for real-world deployment
- Balancing performance & constraints
- Aligning with business objectives

No One-Size-Fits-All Model Why?

- AIoT environments vary widely in:
 - ✓ devices
 - ✓ connectivity
 - ✓ data volume
 - ✓ operational criticality
- Model selection must be context-driven.

Key Factors Influencing Model Choice



**Data availability
& quality**



**Latency
requirements**



**Compute &
memory
constraints**



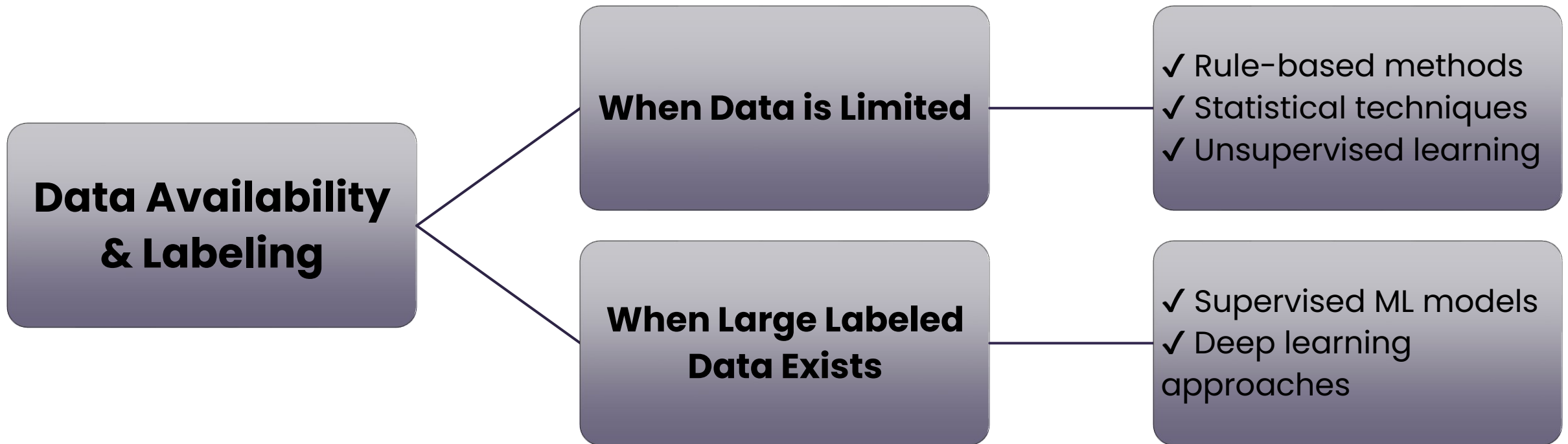
**Network
connectivity**



**Business
priorities**

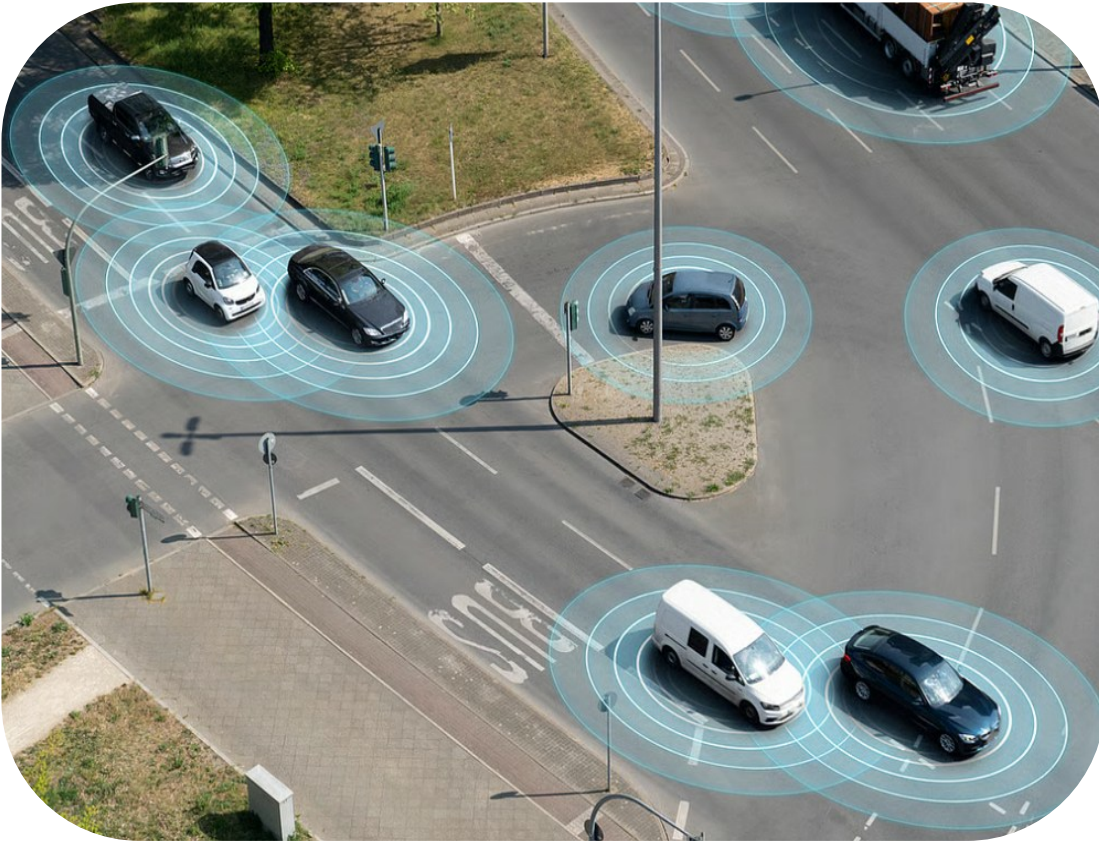
Key Factors Influencing Model Choice

Data Availability & Labeling



Key Factors Influencing Model Choice

Latency & Real-Time Requirements



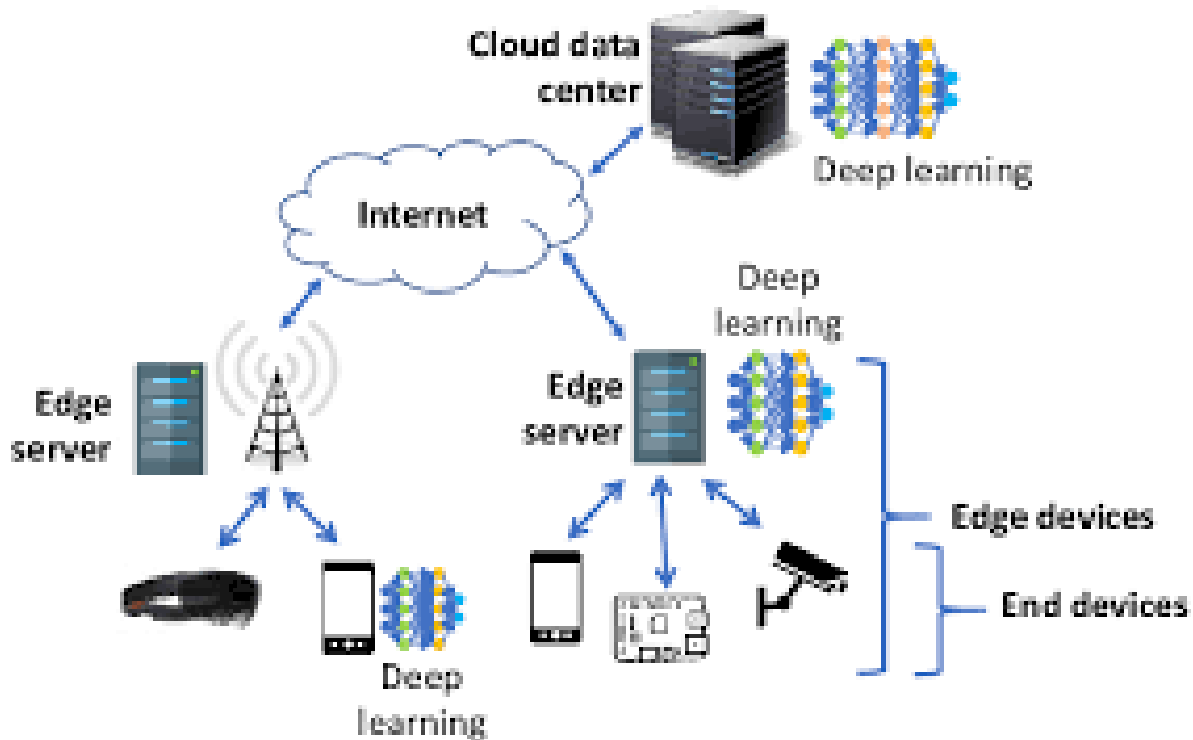
Real-Time AIoT Systems Require

- ⚡ instant decision-making
- ⚡ minimal latency
- ⚡ local processing capability

In safety-critical environments such as robotics, traffic control, and healthcare monitoring, decisions must happen instantly. Lightweight edge models are essential.

Key Factors Influencing Model Choice

Compute & Power Constraints



Edge Device Limitations

- ⚠ low processing capacity
- ⚠ limited memory
- ⚠ battery-powered operation

Suitable Models

- ✓ linear models
- ✓ decision trees
- ✓ lightweight anomaly detection

Edge devices operate with limited processing power and energy capacity. Lightweight models ensure reliable performance and longer device life.

Key Factors Influencing Model Choice

Connectivity & Deployment Environment



Why Connectivity Matters

- 🌐 **Reliable connectivity available**
 - Cloud processing possible
 - Complex models can be used
 - Centralized analytics feasible
- 📶 **Limited or intermittent connectivity**
 - Edge processing required
 - Lightweight models preferred
 - Offline decision-making needed
- 🚫 **No connectivity (remote environments)**
 - Fully autonomous edge intelligence
 - On-device inference essential

In environments with reliable connectivity, data can be processed in the cloud using complex models. However, in remote locations with intermittent or no connectivity, AI decisions must happen locally using lightweight edge models.

Key Factors Influencing Model Choice

Edge vs Cloud Strategy

Edge Models

- ✓ real-time decisions
- ✓ offline operation
- ✓ reliable in low connectivity environments

Cloud Models

- ✓ deeper analytics
- ✓ centralized intelligence
- ✓ require stable connectivity

Hybrid deployments balance both In many AIoT deployments, edge intelligence ensures reliability when connectivity is poor, while cloud systems provide deeper insights when connectivity is available..

Key Factors Influencing Model Choice

Business Goals Drive Model Choice

Model selection must support:



cost reduction



**safety
improvement**



**operational
efficiency**



**regulatory
compliance**



**predictive
insights**

Technology decisions should align with business outcomes. The best model is the one that delivers measurable operational value.

Balancing the Trade-offs

Trade-Off Dimensions

Factor	Simpler Models	Complex Models
Speed	Fast	Slower
Accuracy	Moderate	High
Resource Usage	Low	High
Reliability	High	Variable
Deployment	Edge	Cloud

The goal is not maximum accuracy alone, but the right balance between speed, efficiency, reliability, and cost.

Model Selection Checklist

Ask Before Choosing

- 1 Is real-time response required?
- 2 Can the device support complex models?
- 3 Is labeled data available?
- 4 What is the cost of errors?
- 5 What business outcome is expected?

A structured decision process ensures the selected model meets both operational and business requirements.

Wrap Up – Machine Learning for AIoT

Machine Learning for AIoT – covered

Unsupervised
Learning

Anomaly
Detection

Predictive
Maintenance

Model
Evaluation

Model
Selection

Next Section

- **Deep Learning for AIoT**
- Explore:
 - complex pattern recognition
 - image & audio analytics
 - unstructured data processing